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Scalable information extraction from free text electronic health records using large language models

Bowen Gu^{1,2,3}, Vivian Shao¹, Ziqian Liao³, Valentina Carducci², Santiago Romero Brufau^{2,3}, Jie Yang¹ and Rishi J. Desai^{1*}

Abstract

Background A vast amount of potentially useful information such as description of patient symptoms, family, and social history is recorded as free-text notes in electronic health records (EHRs) but is difficult to reliably extract at scale, limiting their utility in research. This study aims to assess whether an “out of the box” implementation of open-source large language models (LLMs) without any fine-tuning can accurately extract social determinants of health (SDoH) data from free-text clinical notes.

Methods We conducted a cross-sectional study using EHR data from the Mass General Brigham (MGB) system, analyzing free-text notes for SDoH information. We selected a random sample of 200 patients and manually labeled nine SDoH aspects. Eight advanced open-source LLMs were evaluated against a baseline pattern-matching model. Two human reviewers provided the manual labels, achieving 93% inter-annotator agreement. LLM performance was assessed using accuracy metrics for overall, mentioned, and non-mentioned SDoH, and macro F1 scores.

Results LLMs outperformed the baseline pattern-matching approach, particularly for explicitly mentioned SDoH, achieving up to 40% higher Accuracy_{mentioned}. openchat_3.5 was the best-performing model, surpassing the baseline in overall accuracy across all nine SDoH aspects. The refined pipeline with prompt engineering reduced hallucinations and improved accuracy.

Conclusions Open-source LLMs are effective and scalable tools for extracting SDoH from unstructured EHRs, surpassing traditional pattern-matching methods. Further refinement and domain-specific training could enhance their utility in clinical research and predictive analytics, improving healthcare outcomes and addressing health disparities.

Keywords Social determinants of health (SDoH), Electronic health records (EHR), Natural Language Processing (NLP), Large language models (LLMs), Clinical information extraction

*Correspondence:

Rishi J. Desai
rdesai@bwh.harvard.edu

¹Division of Pharmacoepidemiology and Pharmacoeconomics,
Department of Medicine, Brigham and Women's Hospital, Harvard

Medical School, 1620 Tremont Street, Suite 3030-R, Boston, MA
02120, USA

²Department of Otorhinolaryngology - Head & Neck Surgery, Mayo Clinic,
Rochester, MN, USA

³Department of Biostatistics, Harvard T.H. Chan School of Public Health,
Harvard University, Boston, MA, USA



Background

A vast amount of potentially useful information such as description of patient symptoms, family and social history, is not recorded as structured fields but found in free-text notes from electronic health records (EHRs). [1, 2] Substantial feature engineering is needed to directly extract and analyze the relevant information contained within free-text notes, which is time and resource intensive [3–5]. Therefore, free-text notes have historically been underutilized for research activities [6].

Various Natural Language Processing (NLP) models have been developed to extract clinical concepts from EHR notes automatically. [7, 8] Traditional rule-based NLP approaches are often constrained by their lack of generalizability across different datasets and settings. [9, 10] Machine learning-based NLP models rely heavily on the costly step of manual data annotation for model training, which makes them harder to scale across diverse concepts, particularly in clinical environments where data annotation requires specialized knowledge, cost and confidentiality considerations [11–13]. In contrast, Large Language Models (LLMs), advanced deep learning models that are pre-trained on large volumes of text, provide a scalable alternative for the task of clinical information extraction [14–16]. With their capabilities of zero-shot and few-shot learning, LLMs can extract target clinical information from EHR notes without the need for complex rules creation or extensive data annotation [17–19].

Social determinants of health (SDoH) of patients are the economic and social conditions that influence individual and group differences in health status [20–22]. They are important for assessing and addressing health disparities, and are critical for clinical interventions and research [23–25]. For instance, quantifying the role SDoH in predisposing patients to adverse health outcomes is often of interest to facilitate development of educational and counseling interventions. [26, 27] In epidemiologic investigations, various domains of SDoH are also frequently considered important confounding variables that require adjustment [28–30]. While structured EHRs typically lack explicit recording of SDoH information, it is frequently available in free-text form in notes [31–33]. As variables of universal interest, we selected SDoH as a representative use case where scalable extraction methods like LLMs can be of high value. In the current study, we aimed to assess an “out of the box” implementation of 8 advanced open-source LLMs without any fine-tuning for their capabilities on extracting 9 SDoH from free-text EHRs, and compare their performance with the traditional approach of using basic pattern matching [34].

Methods

Data source

We used EHR data from the Mass General Brigham (MGB) system, which is the largest healthcare system in the state of Massachusetts. The sampling frame included a total of 1.2 million individuals with deterministically linked EHRs from MGB to insurance claims data from Medicare and Medicaid for the period of 2007–2020. For this study, we used free text associated with the patients’ social history documentation from progress notes using regular expression matching. The free text was not de-identified and was thus only available within the MGB firewall. Since the patients’ social documentation is added incrementally over time, we only used the most recent social documentation for each patient. A fictitious example of the social history documentation is provided in the Appendix.

SDoH questions and manual labels

To identify the SDoH that are frequently reported in social documentation, we first performed a manual review of a random sample of 200 patients’ social documentation and summarized 9 aspects of the patient’s SDoH that appeared in >5% of the notes, which includes marital status, number of children, employment status, educational status, lifestyle factors (use of tobacco, alcohol, illicit drugs, exercise), and cohabitation status. These SDoH aspects are specific constructs from the broad aspects of SDoH [35]. To select the 200 random samples, we first searched the patients’ social documentation among all the EHR notes we have access to from the MGB system. Then, we grouped the resulting social documentation by patients’ Medical Record Number and kept only the most recent social documentation for each patient. Of the random sample of the 200 reviewed patients, we split the first 100 patients’ social documentation as the validation set to help formulate specifically engineered prompts via prompt engineering (described below) and the remaining 100 patients’ social documentation as the test set to evaluate performance after the prompt engineering is done. Following the classic evaluation framework of LLM evaluation, we converted the SDoH extraction as a question-answer problem. That is, for each of the 9 SDoH characteristics, given the EHR notes, we designed a question and candidate options along with the note text as the LLM input, and let the LLM select the option from the candidates. The SDoH questions, together with their distribution of candidate options in the validation and the test set, are shown in Supplementary Table S1.

For each of the 200 patients, two human reviewers (B.G and V.S) manually labeled the 9 SDoH aspects to one of the quantified choices according to the labeling criteria documentation (Supplementary: Annotation Guide

section). Each reviewer labeled the 200 patients independently. The inter-annotator agreement was calculated based on the total number of the 1800 questions (9 SDoH questions * 200 patients) that the two annotators agreed before discussion among all the 1800 questions, which was 93%. For the inconsistently annotated cases, the two annotators discussed them in detail and reached a consensus. New criteria were also added to the labeling criteria documentation that addressed the causes of these inconsistencies.

Experiment settings

This study was conducted on a server owned by the Mass General Brigham, Department of Medicine, with 20-core Intel Xeon Gold 6248 CPU, 1 NVIDIA V100 PCIE 32GB VRAM GPU, and 754GB RAM. We selected 8 well performing open source LLMs on the LLM leaderboard hosted by Hugging Face [36–45]. To process this study's free text data that contains PHI, all LLMs used in this study are loaded within the MGB firewall, but the LLMs themselves are publicly available and can be used out of the MGB firewall if used to process publicly available data. The details of the LLMs and the links to the models can be found in Supplementary Table S2. During inference, we set the “max_length” parameter to 2048 tokens to avoid potential out-of-memory issue and kept all the parameters their default values. A copy of the model weights went through the AWQ quantization process to generate the quantized model weights. [46, 47] Quantization is a technique of reducing the model size for faster inference in resource limited settings.

Rule-based Baseline Model

To evaluate the comparative performance of LLMs against a common baseline, we designed a model that

used pattern matching to extract the answers to the SDoH questions from the patients' social documentation. The matching patterns were designed based on general intuition for each concept, which was informed by the phrases frequently used in actual patients' social documentation in our validation set. If a match was found in the patients' social documentation, the output answer was guaranteed to be one of the choices of the SDoH question. If no match was found in the patients' social documentation, the output answer was “Not mentioned”. To avoid mismatching (e.g. answer “No” was matched to “Not mentioned” since “No” was a substring of “Not mentioned”), we sorted the choices by their character lengths in descending order and matched the response from the longest choice to the shortest choice and stopped matching if the longer choice was matched. The specific patterns for each SDoH question are shown in Supplementary Table S3.

Pipeline workflow and prompt engineering

We built two pipelines: a default pipeline and a refined pipeline. The default pipeline was designed to run the LLM to extract the SDoHs from the unstructured social documentation using the default prompt for all SDoH questions. Alternatively, the refined pipeline used different engineered prompts on 3 (Q2, Q6, and Q7) of the 9 SDoH questions that most LLMs struggled with in the validation set experiments. The engineered prompts are generated through iterative manual evaluation of incorrect responses from the LLMs and are designed to address the likely root causes of the incorrect responses. We ran both pipelines to compare the effectiveness of the refinement. An illustration of the two pipelines is shown in Fig. 1. The “LLM Response Postprocessing” step included implementing a systematic code that uses

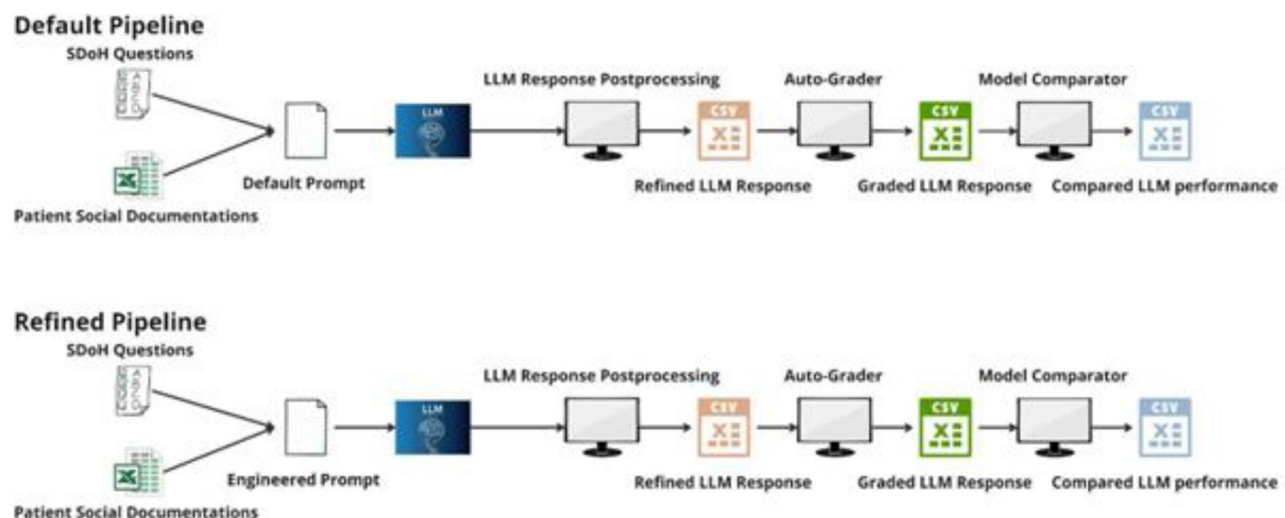


Fig. 1 The default and the refined pipeline

pattern matching to map the model response to one of the choices of the SDoH questions. The “Auto-Grader” took the refined response from the “LLM Response Post-processing” procedure and compared it against manual labels. When mapped model response matched exactly with the human label, then the auto-grader considered it as an accurate extraction. Otherwise, the auto-grader considered the LLM extraction inaccurate. The “Model Comparator” step combined the graded model responses from the “Auto-Grader” and provided the grading results in a single chart to formulate the final benchmark.

Additional context for the “Default Prompt” and the “Engineered Prompt” is shown in Supplementary Table S4, which summarizes 4 types of prompts: default prompts (not including the default secondary prompts), premise prompts, special prompts, and secondary prompts. The default pipeline only used the default prompts while the refined pipeline used all 4 types of prompts. The prompts involved in both pipelines are 0-shot prompts.

LLM performance evaluation

To evaluate the model performance, we used three metrics in the test set: $Accuracy_{overall}$, $Accuracy_{mentioned}$, and $Accuracy_{non-mentioned}$, which corresponds to the overall accuracy for extractions, the accuracy when a note contained mention of the specific SDoH, and the accuracy when a note did not contain mention of the specific SDoH, respectively. The three accuracies are defined as follows, where the $Accuracy_{overall}$ is a weighted average of the $Accuracy_{mentioned}$ and the $Accuracy_{non-mentioned}$ with the weights dependent on the missingness of the SDoH aspects in the text:

$$Accuracy_{overall} = \frac{\text{Total \# of accurate extractions}}{\text{Total \# of questions}}$$

$$Accuracy_{mentioned} = \frac{\text{Total \# of accurate extractions where the ground truth is NOT labeled as “not mentioned”}}{\text{Total \# of questions where the ground truth is NOT labeled as “not mentioned”}}$$

$$Accuracy_{non-mentioned} = \frac{\text{Total \# of accurate extractions where the ground truth is labeled as “not mentioned”}}{\text{Total \# of questions where the ground truth is labeled as “not mentioned”}}$$

We used “Total # of questions” instead of “Total # of valid extractions” as the denominator when calculating the accuracies since it makes the denominator identical for all evaluated LLMs and penalizes LLMs that give invalid responses.

We calculated all three metrics on all 9 SDoH questions. To calculate the confidence interval for these accuracies, we used the Jackknife resampling technique to generate samples for each model accuracy on every question [48]. We then calculated the 95% CI of each accuracy using the samples generated. We assumed t-distribution since our sample size was small (100 per model, per question, per accuracy). For questions that did not meet the premise (e.g. The human label is “not mentioned” for this patient on this SDoH question when trying to calculate $Accuracy_{mentioned}$), we marked them as not applicable for calculations. To evaluate the performance difference between the LLM and the baseline, we also used the Jackknife resampling to generate samples for each model on each question on each accuracy. After that, we performed a two-sided Welch’s t-test on the accuracy differences between the LLM and the baseline using the samples generated from the LLM and the baseline, assuming t-distribution.

Additionally, when the post-processing procedure could not map the LLM response to one of the pre-defined choices, we defined the response to be invalid and reported the proportion of invalid responses for all models. We further reported the F1 score, calculated as a harmonic mean of precision and recall, using the macro averaging method considering all the SDoH questions as multi-class classification problems. When calculating F1 scores for LLM responses, we combined invalid responses with the ‘not mentioned’ category for each question as a default since invalid responses from LLMs lacked a corresponding manual label.

Results

LLM Accuracy

To compare the performance between the LLMs and the baseline, we averaged the model $Accuracy_{overall}$ across the 9 SDoH questions for each LLM and the baseline and did this for the three accuracies defined above (Fig. 2(a) and (b)). The baseline model achieved a 77.33% average $Accuracy_{overall}$, which was mainly attributable to its high average $Accuracy_{non-mentioned}$. Among all the LLMs, openchat_3.5 was the only LLM that had a significantly better average $Accuracy_{overall}$ than the baseline, it was also the only LLM that outperformed the baseline on each of the 9 SDoH questions on the average $Accuracy_{overall}$. On the other hand, the two Llama-2 models had the worst performance on the average $Accuracy_{overall}$ and were significantly lower than the baseline. The remaining LLMs had comparable average $Accuracy_{overall}$ than the baseline, with the zephyr-7b models performing slightly worse. The baseline, which only had limited patterns to match, had an average $Accuracy_{mentioned}$ of 39.02%; while the majority of the LLMs achieved a higher average $Accuracy_{mentioned}$. Openchat_3.5 achieved over 40% higher of the

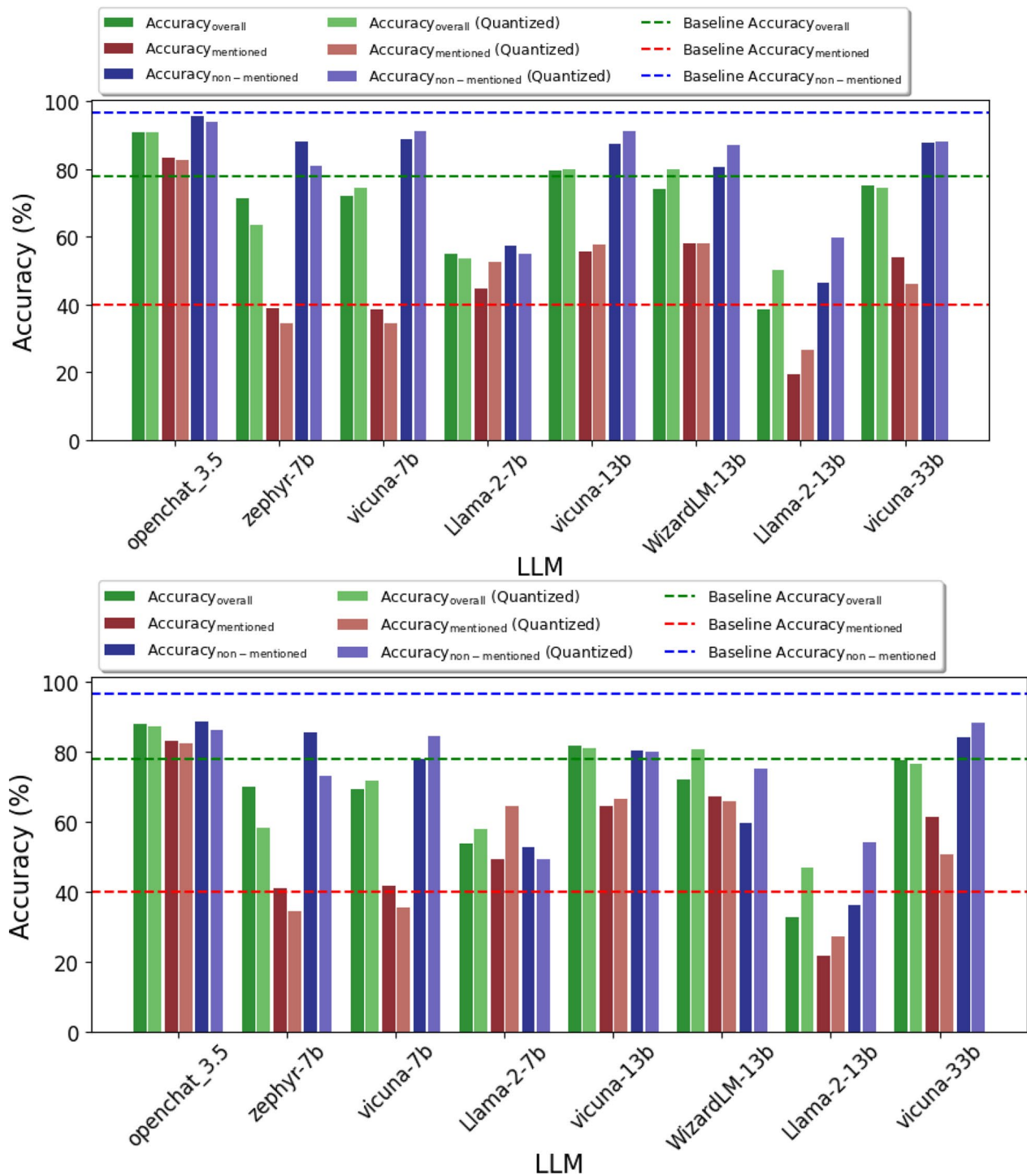


Fig. 2 (a) Average accuracies comparison of the LLMs over the baseline (Refined pipeline). (b) Average accuracies comparison of the LLMs over the baseline (Default pipeline)

average Accuracy_{mentioned} than the baseline, which shows its superior capability in extracting information when it is contained in free text. On the other hand, Llama-2-13b-chat was the only LLM that had a significantly worse average Accuracy_{mentioned} compared to the baseline.

The Accuracy_{non-mentioned} is the accuracy that indicates the model's capability of not producing false information (i.e. avoiding "hallucination") [49–51]. The baseline, which reported its responses using pattern matching, had a very slim probability of producing false responses unless the information recorded in the text was internally

contradicting. On the other hand, the LLMs had a higher probability of producing false information. We found that the baseline achieved an average Accuracy_{non-mentioned} of 96.66%. This was reasonable as the default response from the baseline is “Not mentioned”. The openchat_3.5 model was the best model in terms of minimizing hallucination. On the contrary, the two models that had poor average Accuracy_{non-mentioned} were the two Llama-2 models.

Compared to the default pipeline, the refined pipeline helped reduce LLM hallucination, but in some instances, this came at a cost of reduced sensitivity to the SDoH features mentioned in the text. This is expected as we used the premise prompt in the refined pipeline, which aims at reducing model hallucinations, but this prompt can make the model more conservative in its responses.

We observed that among all the 9 SDoH questions, Q2 (How many children does the patient have?) and Q7 (What is the patient’s employment status?) were the two questions that most LLMs (except the two Llama 2 models) had a better Accuracy_{overall} than the baseline on both the default and the refined pipeline (Supplementary Table S6-S7). This can be explained by Supplementary Table S8-S9, which shows that the Accuracy_{mentioned} for the baseline on Q2 and Q7 were very low compared to the LLMs. Besides Q2, and Q7, Supplementary Table S8-S9 indicates that the baseline model also had very poor Accuracy_{mentioned} on SDoH questions Q3 (Does the patient currently use tobacco?) and Q5 (Does the patient currently use illicit drugs?). This is likely attributable to

the fact that the unstructured social history documentation has many ways of expressing the answers on these questions, a simpler rule-based approach misses most of them. Supplementary Table S10-S11 indicates that the baseline had a much lower probability to produce false positive results compared to LLMs among all 9 SDoH questions. This eliminated the gains in accuracy by LLMs on the Accuracy_{mentioned} and makes the advantage of most LLMs (except the openchat_3.5 model) slim or even negative on the Accuracy_{overall}. For both pipelines, we did not find cases where LLMs correctly identified an SDoH aspect that was missed by the human reviewers.

Macro F1 score

We averaged the model macro F1 scores across the 9 SDoH questions for each LLM and the baseline (Fig. 3). The baseline had 0.53 average macro F1 for the default pipeline and 0.54 average macro F1 for the refined pipeline. Among all the models, openchat_3.5 was the only LLM that had a clear advantage over the baseline. The two Vicuna models and the WizardLM model had comparable performance as the baseline. The zephyr model and the two Llama 2 models had worse average macro F1 scores than the baseline.

Invalid responses

Figure 4 shows the comparison of the invalid responses between the refined and the default pipeline. It indicates that the refined pipeline greatly reduced the number of

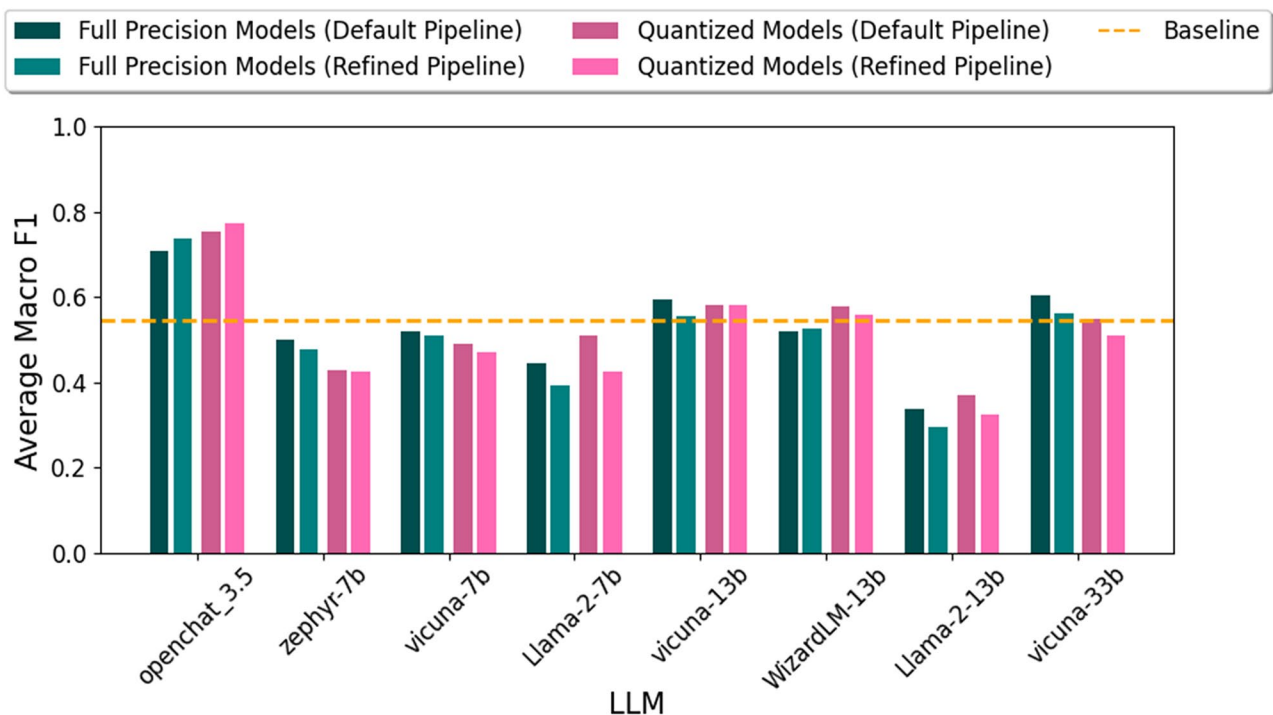


Fig. 3 Average Macro F1 score of LLMs

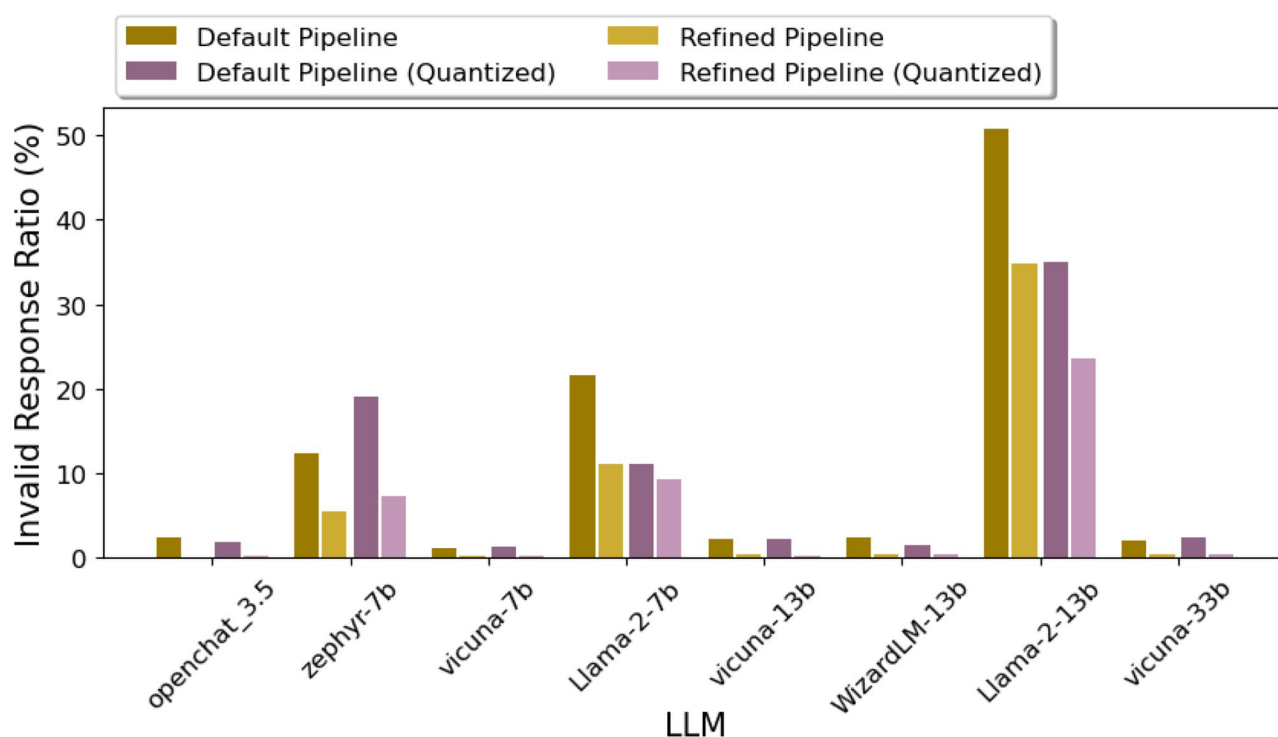


Fig. 4 Invalid response rate comparison between the default and the refined pipeline

invalid responses for each LLM. Some examples of the models' invalid responses and the corresponding analysis are shown in Supplementary Table S12.

High error rate cases

To have a deeper understanding of the cases that most LLMs did poorly, we manually reviewed some high error rate cases for each SDoH question and made comments about how the human label is deduced and the potential reason that the model gets it incorrectly. For all the cases reviewed, at least 75% of the LLMs responded with an incorrect answer. The SDoH question, the text, the human labels, the LLM response distribution, and the human comments are shown in Table S13. Due to HIPAA restrictions, the text shown in the table has been modified with their PHI data removed.

Discussion

In this work, we designed a pipeline that used various open-sourced LLMs to extract patients' SDoH based on social history documentation in free text EHRs. Using prompt engineering and postprocessing, we demonstrated that some LLMs achieved better performance than the baseline approach of pattern matching on most of the SDoH questions, especially when the answer to the SDoH question was mentioned in the text. We also observed that quantization, which could make these models more applicable in resource limited settings,

did not substantially compromise the performance on extraction tasks.

Our study offers some practical learnings regarding using LLMs as scalable extraction tools from free text EHRs. First, thoughtful prompt engineering, as we conducted in our 'refined' pipeline, can substantially elevate model performance, and should be routinely considered. Second, manual review of erroneous model responses indicated that in almost all such instances, the text was either controversial or required strong deductive reasoning to infer the correct answer, which is a known challenge for LLMs [52–54]. We also found these cases had a high overlap with the cases reported by the human reviewers as ambiguous or controversial, and the reviewers needed to expand the labeling criteria to reach an agreement on such cases, which showed that such cases are also challenging to humans. Third, we documented clear and large differences between LLMs in this task of information extraction, which underscores the importance of evaluating multiple models when considering specific tasks. Our findings suggesting robust performance of "out of the box" implementation of LLMs without any fine-tuning are complementary to that of a prior report [16].

The use of LLM pipelines for generating structured SDoH data from unstructured EHR notes could facilitate numerous downstream applications. One significant area is predictive analytics, where structured SDoH information can be routinely considered alongside structured

clinical information to improve risk prediction for various health outcomes. This capability is particularly valuable in managing chronic diseases, where social and environmental factors, such as income and community, play a crucial role. Additionally, structured SDoH data can enhance population health management by enabling healthcare providers to identify and address social disparities in health. In clinical research, the availability of structured SDoH data can refine the selection of study cohorts, improve confounding adjustment in epidemiologic studies, and improve recruitment efforts in clinical trials that aim to oversample socioeconomically disadvantaged patient populations.

The strengths of this work include the automation of the SDoH extraction process with high accuracy. As LLMs avoid the need for model training, this approach is potentially scalable and transportable across institutions with requirement for limited manual effort that can be reserved for validation. Limitations include the complexity and subjectivity involved in the prompt engineering and LLM response postprocessing steps. Additionally, a BERT-based baseline model is not used due to insufficient training data. Also, only one inference cycle is done due to computational resource limitation. Moreover, for implementation simplicity and due to the intrinsic difficulty of LLMs in following the prompt instructions, no mechanism is implemented to enforce a structured output, which potentially increases LLMs' invalid response rate (for example, the Llama 2 models). Furthermore, the matching patterns of the baseline models are designed by gathering the frequently appeared phrases that indicate the answers from the validation set, which may not be comprehensive enough to catch all cases in the test set even if the corresponding information is presented. Besides, the combining of invalid responses with the "not mentioned" category for the SDoH questions when calculating the macro F1 scores limits the ability to differentiate model performance on these two distinct responses. Further, as most open source LLMs are trained using non-clinical text data, specific training on free-text EHRs maybe needed for extraction of more complex clinical concepts with LLMs and our observations regarding model performance and scalability across SDoH domains may not extrapolate to other clinical feature extraction tasks.

Conclusions

In conclusion, we demonstrated the feasibility of employing open weight LLMs to extract patients' SDoH with high accuracy without any additional finetuning. We also demonstrated that prompt engineering was able to reduce invalid response rate for the LLMs and quantized models has relatively robust performance on this task, which could be important for resource limited settings.

LLMs can offer effective and efficient information extraction on various SDoH domains from EHR text.

Abbreviations

AWQ	Activation-aware Weight Quantization
EHR	Electronic Health Records
LLM	Large Language Model
MGB	Mass General Brigham
NLP	Natural Language Processing
SDoH	Social Determinants of Health

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12874-025-02470-z>.

Supplementary Material 1

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Not applicable.

Author contributions

BG and VS processed the data, labeled the data, analyzed the data, and drafted the manuscript. ZL contributed the idea of the secondary prompt, which significantly improved the study results. VC and SB contributed the idea of the pipelines in this study. JY and RD proposed the idea of the study, supervised the study, and refined the manuscript. All authors read and approved the final manuscript.

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Data availability

The data that support the findings of this study are available from Mass General Brigham (MGB) but restrictions apply to the availability of these data, which were used under license for the current study to protect patient privacy, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of Mass General Brigham (MGB). The implementation of the pipelines is made publicly available at <https://github.com/gubowen2/Scalable-information-extraction-from-free-text-electronic-health-records-using-large-language-models>.

Declarations

Ethics approval and consent to participate

Mass General Brigham (MGB) Institutional Review Board approved the study protocol (2020P001486). Consent to participate was not deemed required for this observational investigation.

Consent for publication

Not applicable.

Competing interests

Dr. Desai reports serving as Principal Investigator on investigator-initiated grants to the Brigham and Women's Hospital from Novartis, Vertex, and Bayer on unrelated projects. Other authors do not have any competing interests to disclose. Other authors do not have any competing interests to disclose.

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